

Recitation 12: Pittsburgh Supercomputing Cluster

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Carnegie
Mellon
University



University of
Pittsburgh

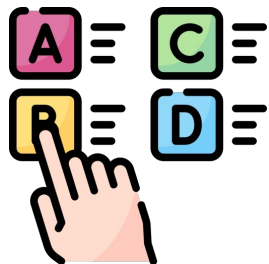
Pittsburgh Supercomputing Cluster: No more GPU-shy!

A bit of trivia

Founded in 1986

Joint venture between CMU & UPitt

Supported by many such as NSF,
federal agencies and industry



It's a computational research center

Simply put, it's a HUGE computer!

Train large-scale research projects

Multiple GPUs and large storage;
undertake large projects



How can it help you: Grants

Empowering the next gen of research

Provides grants and resources

From a few hundred to few thousand
GPU hours of grants available



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PSC User Policy

1. Please only use **V-100** GPUs.
2. Do not download datasets or install environments locally — use shared resources.
3. For your **homework**, always work under **/jet/home/username** directory which has 25 GB of storage.

For your project, the directory will be: `/ocean/projects/<project id>/<username>`, which we will introduce in the next recitation:)

PSC Connection via Terminal

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Connect via Terminal

Steps to login:

You'll need 2 Terminal shells (the ones you need to work with are **green**)

The parts of commands marked with **grey** are ones you'll need to edit as per your details

Step	Terminal 1	Terminal 2
1	<code>ssh <your_username>@bridges2.psc.edu</code>	
2	<code>cd /jet/home/<your_username></code>	
3 (IMP)	<code>interact -p GPU-shared --gres=gpu:v100-32:1 -t 1:00:00 -A cis250019p</code>	
4	<code>module load AI/pytorch_23.02-1.13.1-py3</code>	
5	<code>conda deactivate</code>	
6	<code>conda activate <path_to_your_virtual_environment></code>	
7	<code>hostname</code>	
8	<code>jupyter notebook --no-browser --ip=0.0.0.0</code>	
9 (IMP)	<i>(this terminal is now blocked but DO NOT CLOSE IT)</i>	<code>ssh -L 8888:<hostname>:8888 bridges2.psc.edu -l <your_username></code>
		<i>(Run any command you want here)</i>

Please watch the video for more details :)

PSC Connection via VSCode

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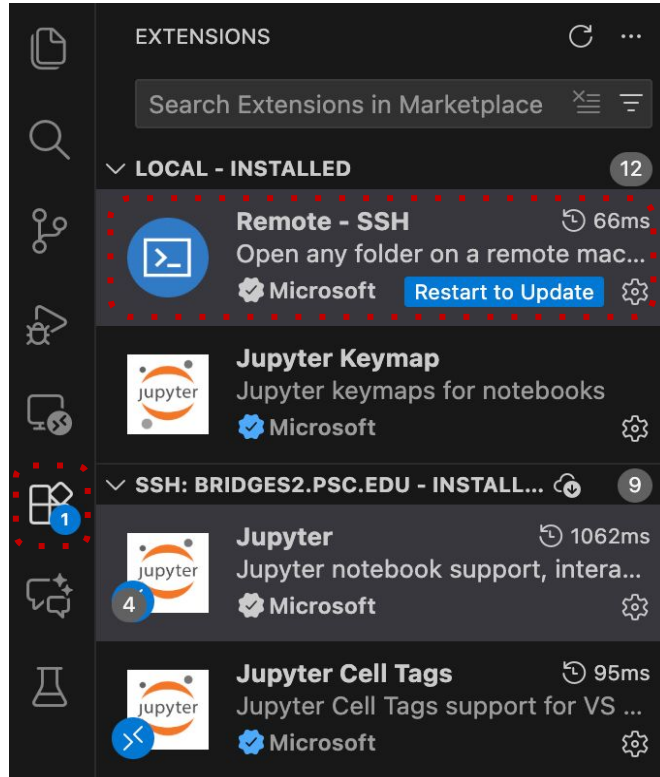
PSC tips

PSC Connection via VSCode

- An Useful link:
- <https://www.psc.edu/resources/bridges-2/user-guide/#vscode>
- It provides a more direct way of connecting using modifying config files.

Connect via VSCode

Please watch the video for more details :)

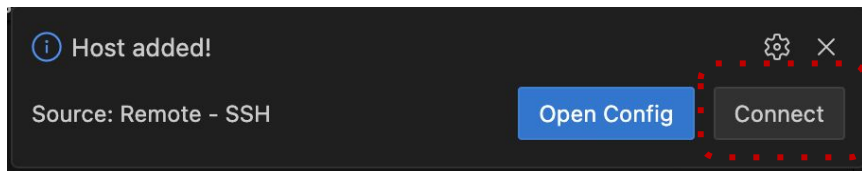
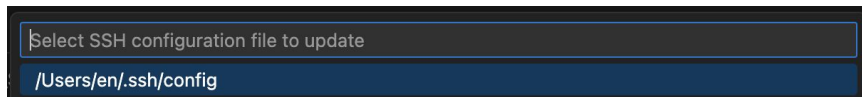
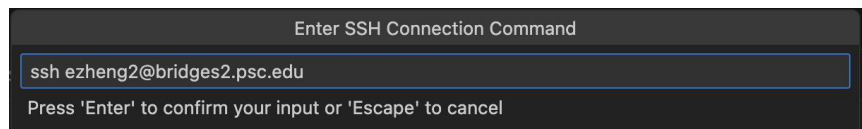
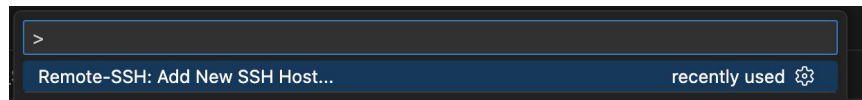


1. SSH into Bridges2

1) Open VS Code and click on the `Extensions` icon in the left sidebar. **Make sure the "***Remote - SSH***" extension is installed.**

Connect via VSCode

Please watch the video for more details :)



2) Open the command palette (**Shift+Command+P** on Mac, **Ctrl+Shift+P** on Windows). A search box will appear at the top center. Choose “Remote-SSH: Add New SSH Host”, then enter:

```
ssh <your_username>@bridges2.psc.edu
```

Next, choose “/Users/<your_username>/.ssh/config” as the config file.

A dialog will appear in the bottom right saying “Host Added”. Click “Connect”, and then enter your password.

Connect via VSCode

SSH keygen and config files:

- Visual Studio Code Tutorials: Remote development over SSH
- Remote - SSH - Visual Studio Marketplace

Config Files would look like this:(ideally, you don't need to generate this file yourself, the plugin would take care of it. Just use Add New SSH Host and input the command!)

In C:/Users/<username>/ssh/config (windows for example)

Host bridges2.psc.edu

HostName bridges2.psc.edu

User <your user name>

Please watch the video for more details :)

Connect via VSCode

Please watch the video for more details :)

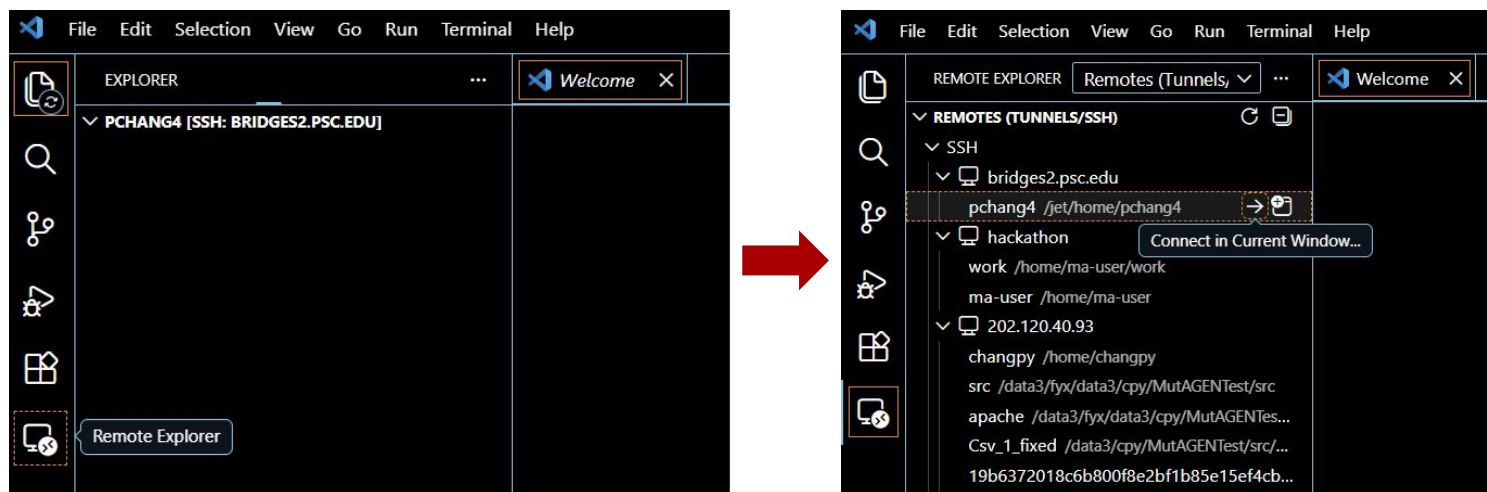
Trouble shooting for SSH connection in VSCode:

1. Try ssh command in bash terminal (For Windows Users). If ssh isn't found, install **OpenSSH Client** (Windows Settings → Optional features)
2. Make sure you have the correct username and password (which might not be the same as ACCESS ID & password and of course not the same as Andrew ID & password).

Connect via VSCode

Please watch the video for more details :)

Useful shortcut for connection:



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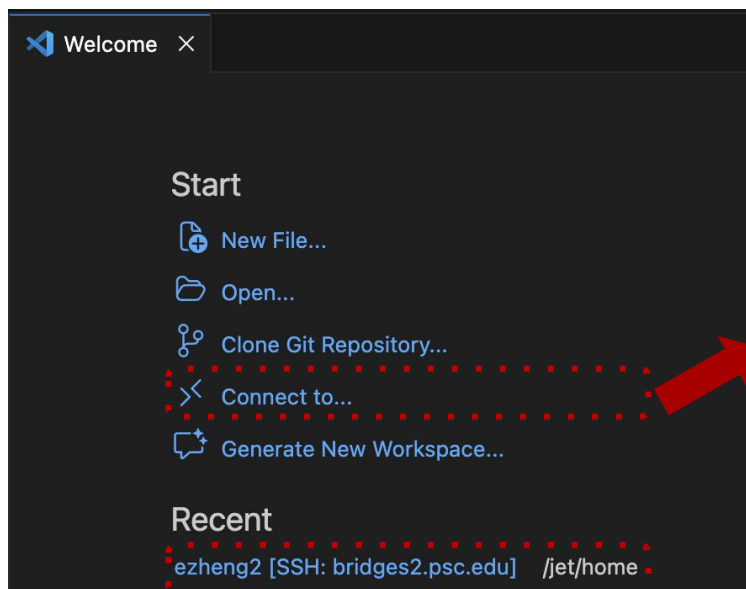
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Please watch the video for more details :)

Also, you can use "Remote-SSH: Connect to Host" and select "bridges2.psc.edu" from the list after adding the host once.



Select an option to open a Remote Window

Connect to Host...

Remote-SSH

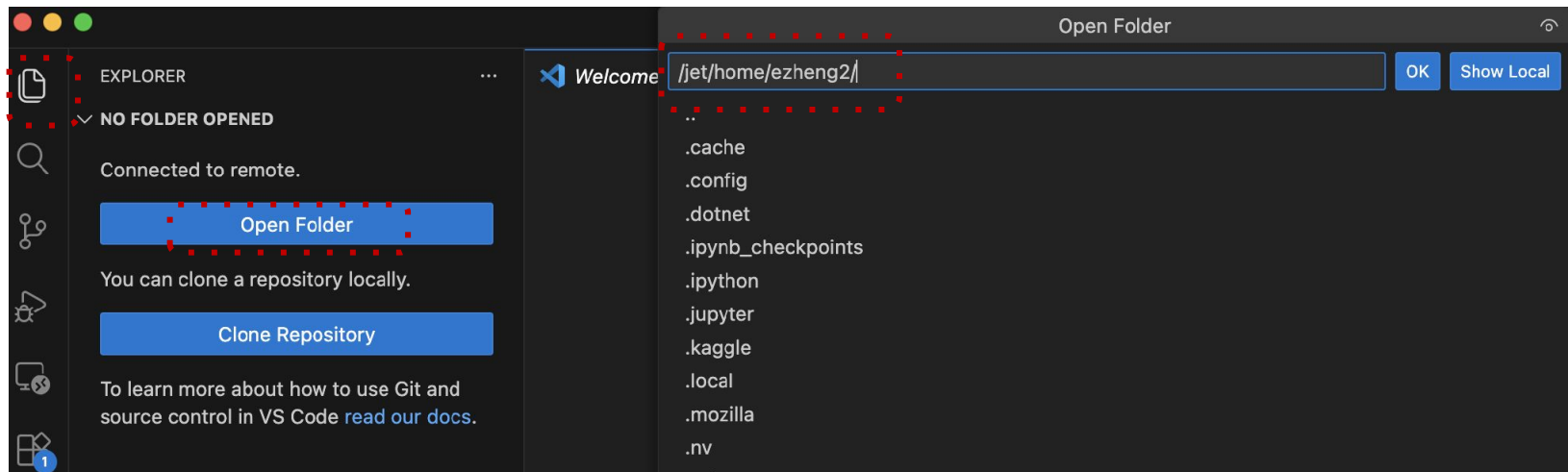
Select configured SSH host or enter user@host

bridges2.psc.edu

Connect via VSCode

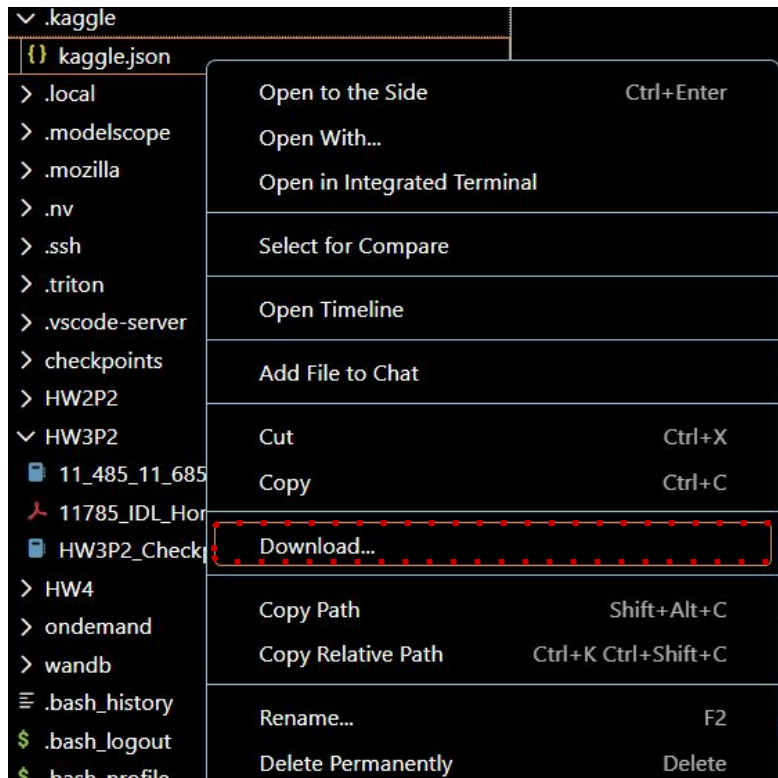
Please watch the video for more details :)

3) Once connected, click “Explorer” in the left sidebar > “Open Folder”, and navigate to your home directory under the project grant:
`/jet/home/<your_username>`



Connect via VSCode

Please watch the video for more details :)



4) You can now **drag** your notebook files directly into the right-hand pane (your remote home directory), or **upload** them using 'scp' into your folder.

Right click the file(Mouse2), you can **download** your file from PSC, which is useful when you try to handin your code.

Connect via VSCode

Please watch the video for more details :)

***You can simply enter the following commands which are given for each HWs in the terminal for operation.**

2. Navigate to Your Directory

`cd /jet/home/<your_username>` #change <your_username> to your username

3. Request a Compute Node

`interact -p GPU-shared --gres=gpu:v100-32:1 -t 8:00:00 -A cis250019p`

4. Load the Anaconda Module

`module load anaconda3`

5. Activate the provided HW Environment**

`conda deactivate` # First, deactivate any existing Conda environment

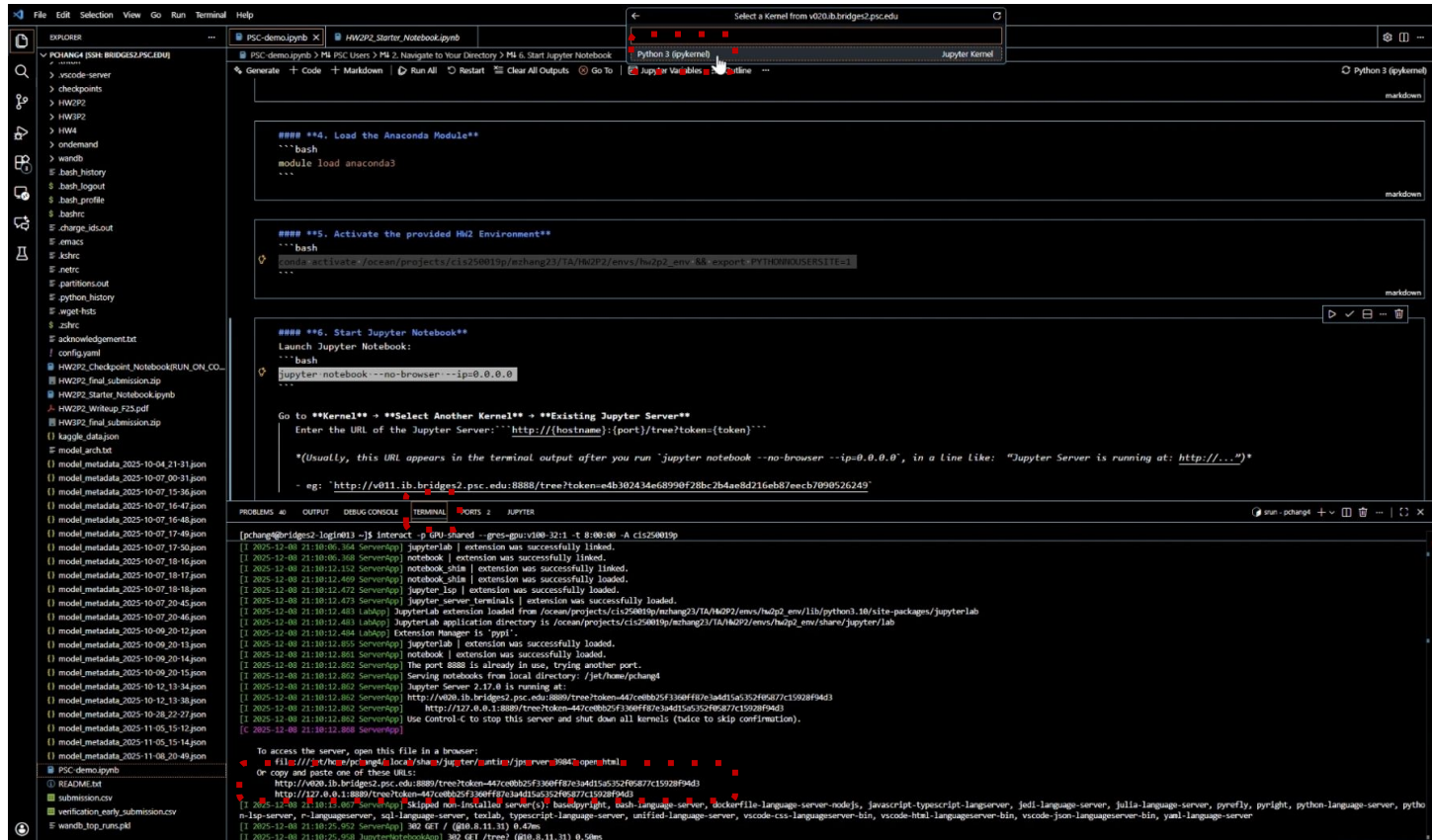
`conda activate /ocean/projects/cis250019p/mzhang23/TA/HW1P2/envs/hw1p2_env && export PYTHONNOUSERSITE=1`

6. Start Jupyter Notebook

`jupyter notebook --no-browser --ip=0.0.0.0`

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PSC Tips

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PSC tips

```
[ezheng2@v019 ~]$ squeue -u $USER:
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(Reason)
36511503	GPU-share	Interact	ezheng2	R	0:14	1	v019

- To check your running jobs, use:
squeue -u \$USER

This shows your current job status. The columns mean:

JOBID: your job ID (unique number)

PARTITION: which queue you are in (e.g. GPU-shared)

NAME: job name (e.g. Interact)

USER: your username

ST: status (R = running, PD = pending, etc.)

TIME: how long the job has been running

NODELIST: which compute node you are on (e.g. v002)

PSC tips

```
[ezheng2@v019 ~]$ squeue -u $USER
      JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
      36511503 GPU-share Interact  ezheng2  R        0:14        1 v019
[ezheng2@v019 ~]$ ssh v019
***** W A R N I N G *****
You have connected to v019.ib.bridges2.psc.edu, a gpu node of Bridges 2.
```

-If you **get disconnected** (e.g. network drop, terminal crash), your GPU job is still running in the background. Since you cannot start another GPU job before finishing the previous one, you can reattach by:

```
ssh <node>
```

For example, if squeue shows v019, then run ssh v019.

PSC tips

```
[ezheng2@v019 ~]$ squeue -u $USER
  JOBID PARTITION   NAME       USER ST       TIME  NODES NODELIST(REASON)
   36511503 GPU-share Interact ezheng2  R       6:41      1 v019
[ezheng2@v019 ~]$ scancel 36511503
salloc: Job allocation 36511503 has been revoked.
[ezheng2@v019 ~]$ srun: Job step aborted: Waiting up to 402 seconds for job step to finish.
slurmstepd: error: *** STEP 36511503.interactive ON v019 CANCELLED AT 2025-12-12T10:55:52 ***
Connection to v019 closed.
[ezheng2@v019 ~]$ squeue -u $USER
  JOBID PARTITION   NAME       USER ST       TIME  NODES NODELIST(REASON)
   36511503 GPU-share Interact ezheng2  CG       6:50      1 v019
```

- If you are DONE with your work but still have GPU time left, please **release the resource** with:

scancel <JOBID>

PSC tips

If you can't connect with VSCode, it might be because the 25GB personal storage quota is full, and vscode server can't be downloaded on PSC.

What we can do is to use Terminal to connect and clean up the space.

How to connect via Antigravity

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